

ELE456
Lecture Notes

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About these notes

These notes develop the mathematical foundations of robotics, from linear algebra to models of robot motion, sensing, and planning. Each chapter introduces a topic through definitions, explanations, and worked examples. The material builds progressively, with an emphasis on understanding the assumptions behind each calculation and interpreting the resulting relationships.

Preparation. These notes are prepared with AI assistance from Paolo Stegagno's lecture transcripts and board material and reviewed by the instructor. Errors may remain; please report suspected mistakes.

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Chapter 1

Introduction and linear algebra

Robotics brings together mathematical models of motion, measurements from sensors, and algorithms that determine how a robot should act. Describing a robot's position, relating measurements to its surroundings, and planning its motion all require a precise mathematical language. Linear algebra provides many of the tools used to express these relationships.

This chapter introduces matrices and the operations needed to work with them: addition, multiplication, transposition, and inversion. It then examines orthogonal matrices, whose inverses can be obtained simply by transposing. The examples develop two habits that will remain essential throughout the study of robotics: checking dimensions before performing an operation and preserving the order of factors in a matrix product.

1.1 Matrices and dimensions

A *matrix* is an array of numbers arranged in rows and columns. Its dimensions are written as
number of rows $\times$ number of columns.

Consider the matrix

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 2}. \quad (1.1)$$

It has three rows and two columns. In general, a_{ij} denotes the entry in row i , column j ; for example, $a_{21} = 3$ and $a_{32} = 1$.

1.2 Matrix addition

Two matrices can be added only if they have the same dimensions. Their entries are then added in corresponding positions:

$$(A + C)_{ij} = a_{ij} + c_{ij}.$$

Consider two additional matrices,

$$B = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \in \mathbb{R}^{2 \times 2}, \quad C = \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 1 & 0 \end{bmatrix} \in \mathbb{R}^{3 \times 2}.$$

The sum $A + B$ is undefined because the dimensions differ. The sum $A + C$ is defined, and its calculation is

$$A + C = \begin{bmatrix} 1+0 & 2+1 \\ 3+0 & 1+0 \\ 0+1 & 1+0 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 3 & 1 \\ 1 & 1 \end{bmatrix}. \quad (1.2)$$

Addition is commutative when defined: $A + C = C + A$.

1.3 Matrix multiplication

1.3.1 Dimension rule and row-by-column calculation

For matrices $M \in \mathbb{R}^{m \times n}$ and $N \in \mathbb{R}^{n \times p}$, the product MN is defined and has dimensions $m \times p$:

$$\underbrace{(m \times n)(n \times p)}_{\text{inner dimensions must match}} \longrightarrow m \times p.$$

The entry in row i , column j is the sum of the products of corresponding entries of row i of M and column j of N :

$$(MN)_{ij} = \sum_{k=1}^n m_{ik} n_{kj}. \quad (1.3)$$

Matrix multiplication is therefore a row-by-column operation, rather than entrywise multiplication.

For the matrices A and B above, the dimensions are $(3 \times 2)(2 \times 2)$, so AB is a 3×2 matrix:

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \quad (1.4)$$

$$= \begin{bmatrix} 1 \cdot 1 + 2 \cdot 1 & 1 \cdot 2 + 2 \cdot 1 \\ 3 \cdot 1 + 1 \cdot 1 & 3 \cdot 2 + 1 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 1 & 0 \cdot 2 + 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 4 & 7 \\ 1 & 1 \end{bmatrix}. \quad (1.5)$$

1.3.2 The order of the factors matters

Reversing the factors gives the dimensions $(2 \times 2)(3 \times 2)$. The inner dimensions do not match, so BA is undefined, even though AB exists.

Even when both orders are defined, the results need not agree. With

$$E = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix},$$

direct multiplication gives

$$BE = \begin{bmatrix} 7 & 10 \\ 4 & 6 \end{bmatrix}, \quad EB = \begin{bmatrix} 3 & 4 \\ 7 & 10 \end{bmatrix}.$$

Thus $BE \neq EB$. Matrix multiplication is not commutative in general. Some particular matrices do commute, but factors cannot be exchanged without a reason that applies to the matrices in question.

1.4 The transpose

Transposing a matrix interchanges its rows and columns. In entry notation, $(M^T)_{ij} = m_{ji}$. If M has dimensions $m \times n$, then M^T has dimensions $n \times m$. For the running example,

$$A^T = \begin{bmatrix} 1 & 3 & 0 \\ 2 & 1 & 1 \end{bmatrix} \in \mathbb{R}^{2 \times 3}. \quad (1.6)$$

The transpose of a product reverses the order of its factors:

$$\boxed{(AB)^T = B^T A^T.} \quad (1.7)$$

The dimensions provide a quick check. Here $B^T A^T$ has dimensions $(2 \times 2)(2 \times 3)$, producing a 2×3 matrix as required. The incorrect expression $A^T B^T$ would have dimensions $(2 \times 3)(2 \times 2)$ and is undefined.

Using the matrices defined above, this identity can be verified directly:

$$B^T A^T = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 0 \\ 2 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 1 \\ 4 & 7 & 1 \end{bmatrix} = (AB)^T.$$

Dimension checks can rule out an invalid expression, but matching dimensions alone do not prove that an identity is correct.

1.5 Identity matrices and inverses

The identity matrix I_n has ones on its main diagonal and zeros elsewhere. For a square matrix $M \in \mathbb{R}^{n \times n}$, an inverse M^{-1} satisfies

$$MM^{-1} = M^{-1}M = I_n. \quad (1.8)$$

This ordinary, two-sided inverse is defined only for square matrices, and not every square matrix has one. The condition for invertibility is

$$\boxed{\det(M) \neq 0.} \quad (1.9)$$

A square matrix with zero determinant is called *singular*.

1.5.1 A one-by-one example

For $H = [5]$, the inverse is $H^{-1} = [1/5]$, since

$$HH^{-1} = H^{-1}H = [1] = I_1.$$

This is the familiar reciprocal operation for a nonzero scalar.

1.5.2 A diagonal example and its determinant

The second example is

$$G = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad G^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}.$$

Multiplying verifies the inverse:

$$GG^{-1} = \begin{bmatrix} 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 0 + 0 \cdot \frac{1}{2} \\ 0 \cdot 1 + 2 \cdot 0 & 0 \cdot 0 + 2 \cdot \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2.$$

The reverse product also equals I_2 . For a general 2×2 matrix,

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

Hence $\det(G) = 1 \cdot 2 - 0 \cdot 0 = 2 \neq 0$, as required. Reciprocating the diagonal entries works here because G is diagonal with nonzero diagonal entries; matrix inversion is not generally an entrywise reciprocal operation.

1.6 Orthogonal matrices

A real square matrix L is *orthogonal* if its inverse equals its transpose:

$$\boxed{L^{-1} = L^{\mathsf{T}}.} \tag{1.10}$$

Equivalently, $LL^{\mathsf{T}} = L^{\mathsf{T}}L = I$. This gives a practical way to check orthogonality: form the transpose, multiply, and verify the identity matrix. For an orthogonal matrix, an inverse can be obtained by transposing instead of performing a general inverse calculation.

1.6.1 Verifying orthogonality

Consider

$$L = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}, \quad L^{\mathsf{T}} = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}.$$

Their product is

$$LL^{\mathsf{T}} = \begin{bmatrix} \frac{1}{4} + \frac{3}{4} & -\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} & \frac{3}{4} + \frac{1}{4} \end{bmatrix} \tag{1.11}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2. \tag{1.12}$$

Thus L is orthogonal and the displayed L^{T} is its inverse.

1.6.2 The trigonometric pattern

The preceding example belongs to the family

$$Q(\theta) = \begin{bmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{bmatrix}. \quad (1.13)$$

Multiplying by the transpose gives

$$\begin{aligned} Q(\theta)Q(\theta)^\top &= \begin{bmatrix} \sin^2 \theta + \cos^2 \theta & -\sin \theta \cos \theta + \cos \theta \sin \theta \\ -\cos \theta \sin \theta + \sin \theta \cos \theta & \cos^2 \theta + \sin^2 \theta \end{bmatrix} \\ &= I_2. \end{aligned}$$

The identity $\sin^2 \theta + \cos^2 \theta = 1$ makes this family orthogonal for every real θ .

The numerical example corresponds to $L = Q(\pi/6)$, since $\sin(\pi/6) = 1/2$ and $\cos(\pi/6) = \sqrt{3}/2$. An equivalent parameterization uses cosine on the diagonal:

$$P(\varphi) = \begin{bmatrix} \cos \varphi & \sin \varphi \\ -\sin \varphi & \cos \varphi \end{bmatrix}.$$

The relation $Q(\theta) = P(\pi/2 - \theta)$ follows from the complementary-angle identities. Thus the same numerical matrix is $L = P(\pi/3)$. In either parameterization, orthogonality follows from the same trigonometric identity. Such matrices provide a foundation for describing changes of orientation in robotics.

1.7 Key identities

For matrices of compatible dimensions, the operations developed in this chapter can be collected as follows:

Operation or property	Rule
Addition	$(M + N)_{ij} = m_{ij} + n_{ij}$
Multiplication	$(MN)_{ij} = \sum_{k=1}^n m_{ik}n_{kj}$
Transpose	$(M^\top)_{ij} = m_{ji}$
Transpose of a product	$(MN)^\top = N^\top M^\top$
Inverse	$MM^{-1} = M^{-1}M = I$
Invertibility of a square matrix	$\det(M) \neq 0$
Orthogonality	$M^{-1} = M^\top$

In the multiplication formula, n is the number of columns of M and the number of rows of N . Addition requires equal dimensions, whereas a product requires only that these inner dimensions match. Inverse and orthogonality conditions apply to square matrices. These distinctions determine which algebraic manipulations are valid before any numerical calculation begins.

1.8 Additional exercises

Show your calculations and justify each conclusion. Check dimensions before performing an operation. These exercises can be completed by hand.

1. **Dimensions and valid operations.** Let $A \in \mathbb{R}^{3 \times 2}$, $B \in \mathbb{R}^{2 \times 3}$, and $C \in \mathbb{R}^{3 \times 3}$. Determine which expressions are defined and give the dimensions of each valid result:

$$A + B, \quad AB, \quad BA, \quad CA, \quad AC, \quad A^T + C, \quad B + A^T.$$

2. **Addition and multiplication.** Consider

$$M = \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix}, \quad N = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}, \quad v = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

Compute $M + N$, MN , NM , and Mv . Do M and N commute? Verify numerically that $(M + N)v = Mv + Nv$.

3. **Transposing a product.** Let

$$U = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 3 \end{bmatrix}, \quad V = \begin{bmatrix} 1 & 2 \\ 0 & -1 \\ 2 & 0 \end{bmatrix}.$$

Compute UV and $(UV)^T$, then independently compute $V^T U^T$ and compare the results. Is $U^T V^T$ defined? Explain why it cannot equal $(UV)^T$ in this example.

4. **Determinants and inverses.** For

$$D = \begin{bmatrix} 4 & 0 \\ 0 & -2 \end{bmatrix}, \quad F = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \quad K = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix},$$

use determinants to decide which matrices are invertible. Find D^{-1} . Verify that $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$ is the inverse of K by multiplying in both orders.

5. **Testing orthogonality.** Consider

$$R = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix}, \quad S = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad T = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Test each matrix for orthogonality using its transpose. For each orthogonal matrix, write its inverse. Is an invertible matrix necessarily orthogonal? Support your answer with one of these examples.

6. **A product of orthogonal matrices.** Suppose P and Q are real orthogonal $n \times n$ matrices. Use the transpose-of-a-product rule to show that $(PQ)(PQ)^T = I_n$. Deduce that PQ is orthogonal and express $(PQ)^{-1}$ in terms of P^T and Q^T , paying attention to their order.

Chapter 2

Reference frames and planar rotations

A robot must relate what its sensors observe to a description of the surrounding world. An obstacle may be directly ahead of the robot, yet lie in a different direction when described on a map. Neither description is ambiguous once the reference frame is specified. The task is to express the same geometric point in different coordinate systems.

This chapter develops that change of coordinates for planar frames with a common origin. Their relative orientation is described by one angle, or equivalently by a rotation matrix. The matrix formulation makes it possible to transform coordinates, reverse a transformation, and compose transformations through several frames using the linear algebra of Chapter 1.

2.1 Coordinates depend on a reference frame

A planar Cartesian reference frame consists of an origin and two perpendicular unit directions, denoted $\mathbf{x}$ and $\mathbf{y}$. A world frame W provides a fixed reference for a map, while a robot frame R is attached to the robot. A sensor can have its own frame S .

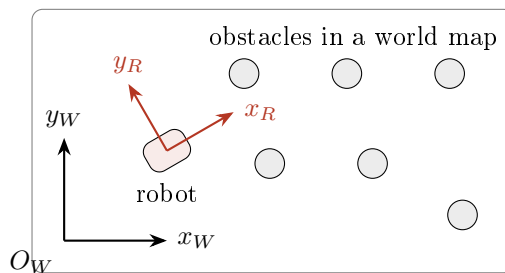


Figure 2.1: A map and a robot use different reference frames. In a general setting their origins and orientations both differ.

If the robot's x_R axis points forward and y_R points left, an obstacle one unit ahead has robot coordinates $[1\ 0]^T$. An obstacle one unit to the left has coordinates $[0\ 1]^T$. These descriptions alone do not specify where the obstacles belong on the world map; the relation between the frames is also needed.

2.1.1 Notation and the common-origin assumption

Let B denote a geometric point. Its coordinates are column vectors,

$$B^W = \begin{bmatrix} x_B^W \\ y_B^W \end{bmatrix}, \quad B^R = \begin{bmatrix} x_B^R \\ y_B^R \end{bmatrix}. \quad (2.1)$$

The superscript states the frame in which the coordinates are expressed. The point B does not change when its coordinate description changes.

For now, assume that W and R share an origin O and have the same orientation convention for their axes. Let $\theta = \theta_R^W$ be the signed angle from x_W to x_R , positive counterclockwise. Only a rotation separates the two frames. Figure 2.2 shows how projections onto their axes produce different coordinates for the same point.

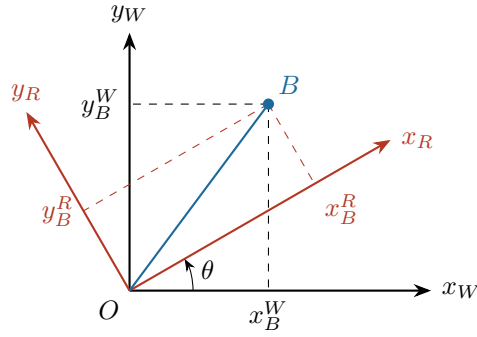


Figure 2.2: A point expressed in two frames with a common origin. Black axes belong to W and red axes to R ; dashed lines give the coordinate projections.

2.2 Constructing the rotation matrix

2.2.1 The columns are the rotated basis vectors

The direction $\mathbf{x}_R$ makes angle θ with x_W . Because it is a unit vector, its components in the world frame are

$$[\mathbf{x}_R]^W = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}. \quad (2.2)$$

The direction $\mathbf{y}_R$ is a further $\pi/2$ counterclockwise, so

$$[\mathbf{y}_R]^W = \begin{bmatrix} \cos(\theta + \pi/2) \\ \sin(\theta + \pi/2) \end{bmatrix} = \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}. \quad (2.3)$$

Here brackets with a superscript mean the components of a geometric vector in the indicated frame. The negative sign in the first component of $[\mathbf{y}_R]^W$ is essential: for a small positive angle, y_R points to the left of y_W .

Putting these two column vectors side by side defines the rotation matrix:

$$R_R^W = [\mathbf{x}_R]^W \quad [\mathbf{y}_R]^W = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}. \quad (2.4)$$

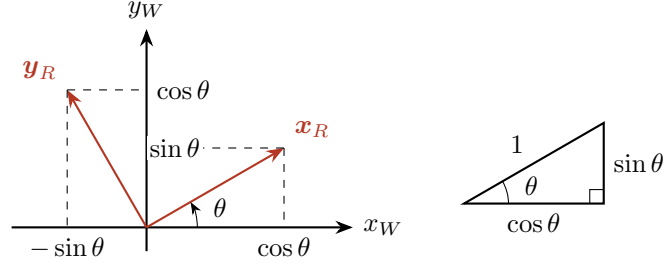


Figure 2.3: Components of the robot's unit basis vectors in the world frame. The right triangle explains the sine and cosine components of $\mathbf{x}_R$.

The subscript identifies the frame whose orientation is being described, and the superscript identifies the reference frame. As a coordinate map, R_R^W takes components expressed in R and returns components expressed in W . An angle is a compact description of orientation; the matrix provides a linear map on coordinate vectors once that angle is fixed.

2.2.2 Orthogonality

Write $c = \cos \theta$ and $s = \sin \theta$. Direct multiplication gives

$$(R_R^W)^\top R_R^W = \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \begin{bmatrix} c & -s \\ s & c \end{bmatrix} = \begin{bmatrix} c^2 + s^2 & 0 \\ 0 & c^2 + s^2 \end{bmatrix} = I_2.$$

Thus R_R^W is orthogonal and

$$(R_R^W)^{-1} = (R_R^W)^\top = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}. \quad (2.5)$$

Its determinant is $c^2 + s^2 = 1$. This last property distinguishes a rotation from an orthogonal reflection, whose determinant is -1 .

2.3 Transforming the coordinates of a point

The vector from the common origin to B can be decomposed along the robot axes:

$$\overrightarrow{OB} = x_B^R \mathbf{x}_R + y_B^R \mathbf{y}_R.$$

Expressing both basis vectors in W gives

$$\boxed{B^W = R_R^W B^R.} \quad (2.6)$$

In scalar form, this is

$$x_B^W = \cos \theta x_B^R - \sin \theta y_B^R, \quad (2.7)$$

$$y_B^W = \sin \theta x_B^R + \cos \theta y_B^R. \quad (2.8)$$

The dimensions are $(2 \times 2)(2 \times 1) = (2 \times 1)$, and the input frame R agrees with the matrix subscript. Both checks are useful: correct matrix dimensions alone do not establish that the intended frames match.

2.3.1 A geometric derivation

For $B \neq O$, let d be the distance from O to B , let α be the angle from x_R to $\overrightarrow{OB}$, and let β be the angle from x_W to that vector. The angles satisfy $\beta = \alpha + \theta$ (modulo 2π). Consequently,

$$x_B^R = d \cos \alpha, \quad y_B^R = d \sin \alpha, \quad x_B^W = d \cos \beta, \quad y_B^W = d \sin \beta.$$

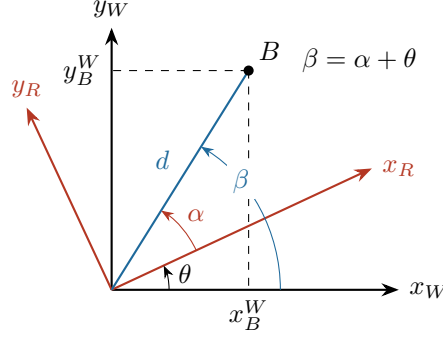


Figure 2.4: The same position vector has angle α in R and angle $\beta = \alpha + \theta$ in W . Its length d is independent of the frame.

Expanding the angle sums gives

$$\begin{aligned} x_B^W &= d \cos(\alpha + \theta) = d \cos \alpha \cos \theta - d \sin \alpha \sin \theta \\ &= \cos \theta x_B^R - \sin \theta y_B^R, \\ y_B^W &= d \sin(\alpha + \theta) = d \cos \alpha \sin \theta + d \sin \alpha \cos \theta \\ &= \sin \theta x_B^R + \cos \theta y_B^R. \end{aligned}$$

These are precisely the entries of (2.6). The matrix formula also applies at $B = O$, where both coordinate vectors are zero and a direction angle is unnecessary.

2.3.2 Worked example: a thirty-degree orientation

Suppose

$$B^R = \begin{bmatrix} 0.8 \\ 0.5 \end{bmatrix}, \quad \theta = \frac{\pi}{6}.$$

Then

$$B^W = \begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} 0.8 \\ 0.5 \end{bmatrix} \tag{2.9}$$

$$= \begin{bmatrix} 0.4\sqrt{3} - 0.25 \\ 0.4 + 0.25\sqrt{3} \end{bmatrix} \approx \begin{bmatrix} 0.4428 \\ 0.8330 \end{bmatrix}. \tag{2.10}$$

The coordinates change, but the point's distance from the common origin does not. Indeed, orthogonality implies

$$(B^W)^\top B^W = (B^R)^\top (R_R^W)^\top R_R^W B^R = (B^R)^\top B^R = 0.8^2 + 0.5^2 = 0.89.$$

This is a change of coordinate description of a fixed point, rather than a physical motion of that point.

2.4 Reversing a coordinate transformation

To recover robot coordinates from world coordinates, multiply Equation (2.6) on the left by the inverse:

$$(R_R^W)^{-1} B^W = (R_R^W)^{-1} R_R^W B^R = B^R.$$

Using orthogonality,

$$\boxed{B^R = (R_R^W)^\top B^W = R_W^R B^W, \quad R_W^R = (R_R^W)^\top.} \quad (2.11)$$

Reversing the frame relation exchanges the frame indices and negates the relative angle. If $R(\theta)$ denotes the matrix in Equation (2.4), then $R(\theta)^\top = R(-\theta)$. For arbitrary frames A and F , the same rule reads

$$R_F^A = (R_A^F)^\top. \quad (2.12)$$

For example, applying the transpose to the exact vector in Equation (2.10) recovers $[0.8 \ 0.5]^\top$.

2.5 Composing rotations through several frames

A measurement may begin in a sensor frame S , pass through a robot frame R , and finally be expressed in W . Assume again that all origins coincide. Let the orientation of R relative to W be θ , and the orientation of S relative to R be φ .

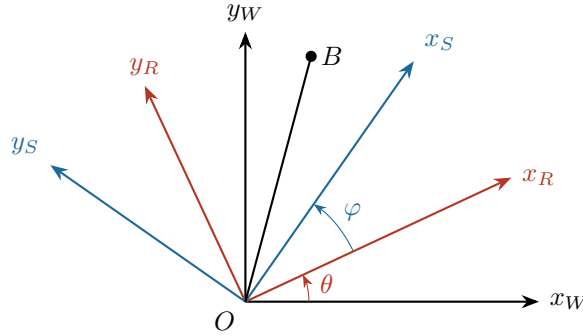


Figure 2.5: Three frames with a common origin. The sensor frame (blue) is rotated by φ relative to the robot frame (red), which is rotated by θ relative to the world frame (black).

The two successive coordinate transformations are

$$B^R = R_S^R B^S, \quad B^W = R_R^W B^R.$$

Substitution gives

$$B^W = R_R^W R_S^R B^S, \quad \boxed{R_S^W = R_R^W R_S^R.} \quad (2.13)$$

The rightmost matrix acts first. Its output is expressed in R , precisely the frame expected by the matrix to its left. In two dimensions,

$$R_R^W = R(\theta), \quad R_S^R = R(\varphi), \quad R_S^W = R(\theta + \varphi),$$

as can be verified with the sine and cosine addition identities.

2.5.1 Reading the frame indices

The general composition rule is

$$R_C^A = R_B^A R_C^B. \quad (2.14)$$

The subscript of the left factor matches the superscript of the next factor. The remaining indices identify the overall output and input frames. This is a rule about the meanings of the maps, not ordinary cancellation of numbers.

For example,

$$R_2^3 R_4^2 R_5^4 R_1^5 = R_1^3. \quad (2.15)$$

Reading from right to left, the coordinates pass through frames $1 \rightarrow 5 \rightarrow 4 \rightarrow 2 \rightarrow 3$. By contrast, in the written product $R_2^3 R_4^2$, the inner frame labels 2 and 4 do not match. Its factors cannot be composed as that sequence of coordinate maps merely by cancelling the repeated 3.

Planar rotation matrices happen to commute numerically: $R(\theta)R(\varphi) = R(\varphi)R(\theta)$. This special property does not change the input and output frame labels of an individual map. Keeping the frame chain explicit also prepares the notation for three-dimensional rotations, where commutativity does not hold in general.

2.5.2 Simplifying chains containing transposes

First apply the transpose rules, then inspect the frame chain. Consider

$$R_2^3 (R_2^4)^\top (R_5^1 R_4^5)^\top.$$

Transposing the last product reverses its factor order, so

$$\begin{aligned} R_2^3 (R_2^4)^\top (R_5^1 R_4^5)^\top &= R_2^3 R_4^2 (R_4^5)^\top (R_5^1)^\top \\ &= R_2^3 R_4^2 R_5^4 R_1^5 \\ &= R_1^3. \end{aligned}$$

Two different operations are involved: a transpose exchanges the frame indices of one rotation matrix, and transposing a product reverses the order of its factors. Both are necessary to obtain a consistent chain.

2.6 The limit of a rotation-only model

The formula $B^W = R_R^W B^R$ describes point coordinates when the origins coincide. In a general robot configuration, the origin O_R is displaced from O_W , as in Figure 2.6. A rotation matrix describes the relative orientation, but contains no information about this displacement.

To see the limitation, take the point $B = O_R$. Its robot coordinates are zero, so $R_R^W B^R = 0$ for every rotation matrix. Its world coordinates, however, are the nonzero displacement of O_R from O_W . A complete point-coordinate transformation must therefore include translation as well as rotation. The rotation rules remain applicable to direction vectors, which do not depend on the choice of origin. Distinguishing point positions from directions is essential when extending the model.

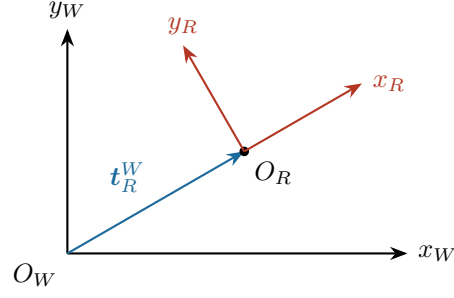


Figure 2.6: Different origins introduce a translation $\mathbf{t}_R^W$. Orientation alone cannot determine the world coordinates of a point measured relative to O_R .

2.7 Key identities

Operation	Identity
Planar rotation	$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$
Point coordinates, common origin	$B^W = R_R^W B^R$
Reverse transformation	$R_W^R = (R_R^W)^\top = R(-\theta)$
Composition	$R_S^W = R_R^W R_S^R$
Planar angle composition	$R(\theta)R(\varphi) = R(\theta + \varphi)$

2.8 Additional exercises

Use column vectors, show your calculations, and label the input and output frames. Unless stated otherwise, all frames share an origin.

1. **Sketching frames and coordinates.** Draw world axes x_W, y_W and robot axes rotated counterclockwise by $\pi/2$. A point has $B^R = [2 \ 1]^\top$. Locate it on your sketch, compute B^W , and check that your answer agrees with its position relative to the world axes.
2. **Constructing a rotation matrix.** The robot frame is rotated by $-\pi/4$ relative to the world frame. Write $[x_R]^W$ and $[y_R]^W$, and use them to construct R_R^W . Verify $(R_R^W)^\top R_R^W = I_2$ and $\det(R_R^W) = 1$.
3. **Forward and reverse transformations.** Let $\theta_R^W = \pi/6$ and $B^R = [1 \ 2]^\top$. Find B^W in exact form. Construct R_W^R and use it to recover B^R . Verify that the squared distance from the common origin is the same in both coordinate descriptions.
4. **From a sensor to the world.** Suppose $\theta_R^W = \pi/6$, $\theta_S^R = \pi/3$, and $B^S = [2 \ -1]^\top$. Compute R_S^W and B^W . Check your result by first finding B^R and then transforming it to W . What is the orientation of S relative to W ?
5. **Frame chains and transposes.** Simplify the first two expressions to a single rotation matrix. For the third, explain why the written frame chain is inconsistent, even though the numerical matrix dimensions permit multiplication:

$$R_B^A R_C^B R_D^C, \quad (R_A^B)^\top (R_C^D R_B^C)^\top, \quad R_B^A R_D^C.$$

Assume distinct letters denote distinct frames. State the input and output frames of each valid chain.

6. **Why an origin matters.** The robot axes are parallel to the world axes, but the robot origin has world coordinates $[3 \ 1]^\top$. A point has robot coordinates $[2 \ 0]^\top$. Use a sketch to find its world coordinates. What would the rotation-only formula predict, and what information would it omit?
7. **Length preservation.** Let $v^W = R_R^W v^R$ be two coordinate descriptions of a direction vector. Using orthogonality, show that $(v^W)^\top v^W = (v^R)^\top v^R$. Explain geometrically why a change of reference frame cannot change the length of the vector.

Chapter 3

Planar rigid transformations and homogeneous coordinates

Rotation describes the relative orientation of two frames, but a mobile robot also changes position. Its origin generally differs from the origin of the world map. A sensor mounted on the robot may have yet another origin and orientation. Relating point coordinates between these frames requires both rotation and translation.

This chapter extends the rotation model of Chapter 2 to planar rigid transformations. The geometric construction leads first to an affine coordinate relationship, then to a homogeneous matrix representation that combines rotation and translation in a single multiplication. The same representation makes composition and inversion systematic.

3.1 Rotation and translation between two frames

Let O_W and O_R be the origins of world frame W and robot frame R . Define

$$p_R^W = \begin{bmatrix} p_{Rx}^W \\ p_{Ry}^W \end{bmatrix}$$

as the position of O_R relative to O_W , expressed in world coordinates. The rotation matrix R_R^W describes the orientation of R relative to W . Together, p_R^W and R_R^W specify the robot frame's planar pose.

For a point B , the geometric displacement vectors obey

$$\overrightarrow{O_W B} = \overrightarrow{O_W O_R} + \overrightarrow{O_R B}. \quad (3.1)$$

This triangle relation is independent of coordinates. To turn it into a numerical calculation, however, every vector being added must be expressed in the same frame.

The column B^R contains the components of $\overrightarrow{O_R B}$ along the robot axes. The column p_R^W uses the world axes. Adding these columns directly would combine components measured along different directions. First rotate B^R into world-aligned components; then add the translation:

$$\boxed{B^W = p_R^W + R_R^W B^R.} \quad (3.2)$$

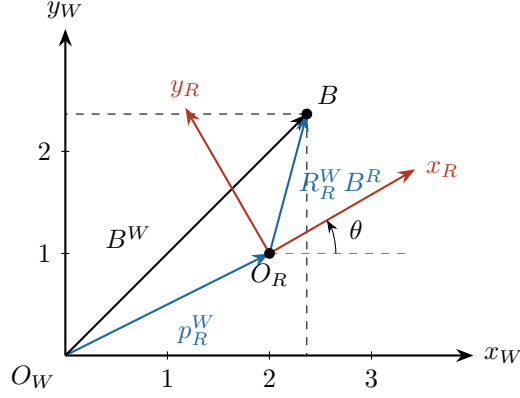


Figure 3.1: The displacement to B is the sum of the displacement to the robot origin and the displacement from that origin to B . All three arrow labels give components in W . The geometry uses $p_R^W = [2 \ 1]^T$, $\theta = \pi/6$, and $B^R = [1 \ 1]^T$.

Here $R_R^W B^R$ describes the displacement from O_R to B using world directions. It is not, by itself, the position of B relative to O_W .

3.1.1 An intermediate frame makes the geometry explicit

Introduce a frame A with origin $O_A = O_R$ and axes parallel to those of W . Frames R and A differ only by rotation, so the previous chapter gives

$$B^A = R_R^A B^R.$$

Since A and W have the same orientation, $R_R^A = R_R^W$. Moreover, the components in A use exactly the directions needed for addition to p_R^W . Thus $B^W = p_R^W + B^A$, recovering Equation (3.2).

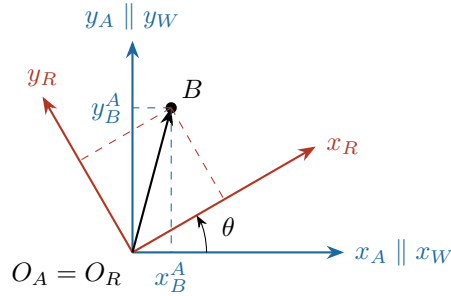


Figure 3.2: The auxiliary frame A shares the robot origin but is aligned with the world. Rotating from R to A resolves the displacement into components that can be added to the world position of O_R .

3.1.2 Why the relationship is affine

A map $f(b) = Rb + p$ is called affine: it consists of a linear map followed by an offset. With a nonzero translation, it is not a linear map of the original two coordinates, since $f(0) = p \neq 0$. In

the present setting R is a rotation, so the transformation is rigid: it preserves distances between points. Distances to the coordinate origin need not be preserved when the origin changes.

3.1.3 Worked example

Suppose the given quantities are

$$B^R = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ m}, \quad p_R^W = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{ m}, \quad \theta_R^W = \frac{\pi}{6}.$$

The rotation matrix is dimensionless. Applying the affine relationship gives

$$B^W = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (3.3)$$

$$= \begin{bmatrix} (3 + \sqrt{3})/2 \\ (3 + \sqrt{3})/2 \end{bmatrix} \approx \begin{bmatrix} 2.3660 \\ 2.3660 \end{bmatrix} \text{ m}. \quad (3.4)$$

The coordinate arithmetic uses metres throughout. The result agrees with Figure 3.1: the point lies above and slightly to the right of the robot origin. Keeping the exact expressions until the final step avoids rounding errors when reversing the calculation.

3.2 Reversing the affine relationship

Suppose B^W is known and B^R is required. Subtract the world-coordinate translation before applying the inverse rotation:

$$B^W - p_R^W = R_R^W B^R, \quad (3.5)$$

$$\boxed{B^R = (R_R^W)^\top (B^W - p_R^W)}. \quad (3.6)$$

The difference in parentheses is expressed in W ; the transpose maps it to R . The parentheses matter: the rotation acts on both terms, not just on B^W . Equivalently,

$$B^R = -(R_R^W)^\top p_R^W + (R_R^W)^\top B^W. \quad (3.7)$$

This has the same affine structure as the forward relationship, with

$$R_W^R = (R_R^W)^\top, \quad p_W^R = -(R_R^W)^\top p_R^W. \quad (3.8)$$

The reversed translation is not generally just $-p_R^W$: it must also be expressed along the axes of the reverse transformation's output frame.

For the exact coordinates in Equation (3.4),

$$B^W - p_R^W = \begin{bmatrix} (\sqrt{3} - 1)/2 \\ (1 + \sqrt{3})/2 \end{bmatrix} \text{ m}.$$

Multiplying by $\begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix}$ recovers $B^R = [1 \ 1]^\top \text{ m}$. If the rounded value $[2.37 \ 2.37]^\top$ is used instead, the recovered result is approximately $[1.0054 \ 1.0015]^\top \text{ m}$, rather than exactly $[1 \ 1]^\top$.

3.3 Composing transformations through a sensor frame

A sensor frame S may be displaced and rotated relative to the robot frame. Given the sensor's pose in R and the robot's pose in W , the two coordinate relationships are

$$B^R = p_S^R + R_S^R B^S, \quad B^W = p_R^W + R_R^W B^R.$$

Substitution yields

$$B^W = p_R^W + R_R^W (p_S^R + R_S^R B^S) \quad (3.9)$$

$$= \underbrace{p_R^W + R_R^W p_S^R}_{p_S^W} + \underbrace{R_R^W R_S^R}_{R_S^W} B^S. \quad (3.10)$$

The overall rotation and translation are therefore

$$\boxed{R_S^W = R_R^W R_S^R, \quad p_S^W = p_R^W + R_R^W p_S^R.} \quad (3.11)$$

The sensor translation must be rotated into W before it can be added to the robot translation. Frame labels explain why the expression is valid and why simply adding $p_R^W + p_S^R$ would be incorrect in general.

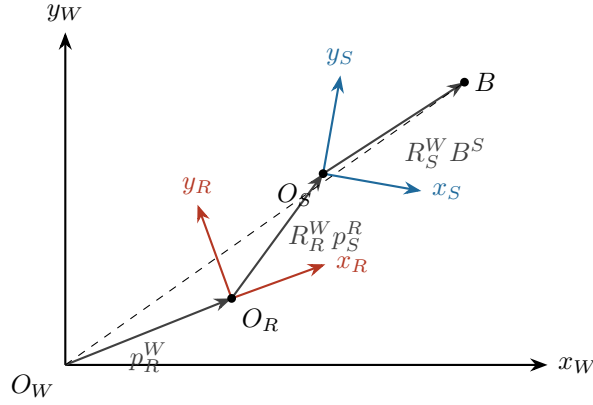


Figure 3.3: A world–robot–sensor chain with distinct origins. Each displacement along the path to B is labeled in world components so that the arrows can be added numerically. The dashed line is the direct displacement to B .

Repeated substitution works for any number of frames, but each additional frame introduces another translation term. A representation that absorbs these additions into matrix multiplication makes longer chains easier to manage.

3.4 Homogeneous coordinates and transformation matrices

3.4.1 Adding one coordinate

For a point with coordinates B^R , define its homogeneous representation by appending a dimensionless one:

$$\tilde{B}^R = \begin{bmatrix} B^R \\ 1 \end{bmatrix} = \begin{bmatrix} x_B^R \\ y_B^R \\ 1 \end{bmatrix}. \quad (3.12)$$

The additional coordinate is bookkeeping for translation, not a third spatial coordinate. Define the planar homogeneous transformation matrix as

$$H_R^W = \begin{bmatrix} R_R^W & p_R^W \\ 0_{1 \times 2} & 1 \end{bmatrix}. \quad (3.13)$$

Its blocks have dimensions 2×2 , 2×1 , 1×2 , and 1×1 , respectively. Written entry by entry, it is

$$H_R^W = \begin{bmatrix} \cos \theta & -\sin \theta & p_{Rx}^W \\ \sin \theta & \cos \theta & p_{Ry}^W \\ 0 & 0 & 1 \end{bmatrix}.$$

The upper-left block contains dimensionless rotation coefficients; the first two entries of the last column carry the chosen length unit. Block divisions are visual aids, not extra mathematical operations.

3.4.2 One product performs the full transformation

Block multiplication gives

$$H_R^W \tilde{B}^R = \begin{bmatrix} R_R^W & p_R^W \\ 0_{1 \times 2} & 1 \end{bmatrix} \begin{bmatrix} B^R \\ 1 \end{bmatrix} = \begin{bmatrix} R_R^W B^R + p_R^W \\ 1 \end{bmatrix} = \tilde{B}^W. \quad (3.14)$$

Thus the affine transformation becomes a single matrix multiplication:

$$\tilde{B}^W = H_R^W \tilde{B}^R. \quad (3.15)$$

The first two output entries are the desired point coordinates. For the normalized point representation used here, the last entry remains one. The fixed bottom row $[0 \ 0 \ 1]$ is what preserves this property.

For the running example, using metres for the position entries,

$$\begin{bmatrix} \sqrt{3}/2 & -1/2 & 2 \\ 1/2 & \sqrt{3}/2 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} (3 + \sqrt{3})/2 \\ (3 + \sqrt{3})/2 \\ 1 \end{bmatrix} \approx \begin{bmatrix} 2.3660 \\ 2.3660 \\ 1 \end{bmatrix}.$$

The physical point coordinates are the first two entries in metres; the final one has no length unit.

3.5 Composition by matrix multiplication

Using homogeneous coordinates for the sensor chain,

$$\tilde{B}^R = H_S^R \tilde{B}^S, \quad \tilde{B}^W = H_R^W \tilde{B}^R.$$

Combining the two maps gives

$$H_S^W = H_R^W H_S^R, \quad \tilde{B}^W = H_R^W H_S^R \tilde{B}^S. \quad (3.16)$$

Multiplying the blocks verifies that this contains both composition rules:

$$H_R^W H_S^R = \begin{bmatrix} R_R^W & p_R^W \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_S^R & p_S^R \\ 0 & 1 \end{bmatrix} \quad (3.17)$$

$$= \begin{bmatrix} R_R^W R_S^R & p_R^W + R_R^W p_S^R \\ 0 & 1 \end{bmatrix} = H_S^W. \quad (3.18)$$

Here each 0 in the bottom-left block denotes a 1×2 zero row. The product is another homogeneous rigid transformation with bottom row $[0 \ 0 \ 1]$. As before, the rightmost map acts first and adjacent frame indices must match. For example,

$$H_1^2 H_3^1 H_4^3 H_5^4 = H_5^2$$

maps coordinates through $5 \rightarrow 4 \rightarrow 3 \rightarrow 1 \rightarrow 2$.

Unlike pure planar rotations, planar transformations with translation do not generally commute. The term $R_R^W p_S^R$ shows why: a preceding rotation changes the direction in which a subsequent local translation acts.

3.6 Inverting a homogeneous transformation

3.6.1 Why the transpose is insufficient

Expanding the determinant along the bottom row gives

$$\det(H_R^W) = \det(R_R^W) = 1.$$

Every matrix of this rigid-transformation form is therefore invertible. However, it is generally not orthogonal. If its translation is nonzero, its transpose has bottom row $[(p_R^W)^\top \ 1]$ rather than $[0 \ 0 \ 1]$. Consequently, transposing the full matrix does not produce its inverse. When the translation is zero, the full matrix is orthogonal and the transpose does coincide with the inverse.

3.6.2 Deriving the inverse from the affine relationship

Equation (3.7) already identifies the reverse rotation and translation. Place these in their corresponding blocks:

$$H_W^R = (H_R^W)^{-1} = \begin{bmatrix} (R_R^W)^\top & -(R_R^W)^\top p_R^W \\ 0_{1 \times 2} & 1 \end{bmatrix}. \quad (3.19)$$

This gives $\tilde{B}^R = H_W^R \tilde{B}^W$ without a general-purpose 3×3 inversion. Transpose the rotation block, compute the rotated negative translation, and retain the bottom row.

For a compact verification, write $R = R_R^W$ and $p = p_R^W$. Then

$$\begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R^\top & -R^\top p \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} R R^\top & -R R^\top p + p \\ 0 & 1 \end{bmatrix} = I_3.$$

The reverse product also equals I_3 , since its translation block is $R^\top p - R^\top p = 0$.

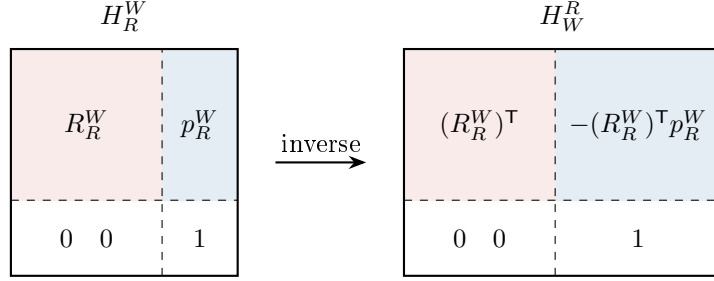


Figure 3.4: Block structure of a transform and its inverse. Red blocks contain rotation; blue blocks contain translation. The inverse translation must be rotated as well as negated. Block widths are schematic.

3.6.3 Worked inverse

For $p_R^W = [2 \ 1]^T$ m and $\theta_R^W = \pi/6$,

$$\begin{aligned} -(R_R^W)^T p_R^W &= - \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} -\sqrt{3} - 1/2 \\ 1 - \sqrt{3}/2 \end{bmatrix} \approx \begin{bmatrix} -2.2321 \\ 0.1340 \end{bmatrix} \text{ m.} \end{aligned}$$

Hence, with translation entries in metres,

$$H_W^R = \begin{bmatrix} \sqrt{3}/2 & 1/2 & -\sqrt{3} - 1/2 \\ -1/2 & \sqrt{3}/2 & 1 - \sqrt{3}/2 \\ 0 & 0 & 1 \end{bmatrix}. \quad (3.20)$$

Its last column locates the world origin in robot coordinates. Its upper-left block describes the world axes in the robot frame. These meanings provide geometric checks on the inverse, in addition to verifying the matrix product.

3.7 Key identities

Operation	Identity
Forward point transformation	$B^W = p_R^W + R_R^W B^R$
Reverse point transformation	$B^R = (R_R^W)^T (B^W - p_R^W)$
Homogeneous point transformation	$\tilde{B}^W = H_R^W \tilde{B}^R$
Composition	$H_S^W = H_R^W H_S^R$
Reverse rotation	$R_W^R = (R_R^W)^T$
Reverse translation	$p_W^R = -(R_R^W)^T p_R^W$

Always specify the frame of a translation, express displacements in a common frame before adding them, and distinguish the inverse of a rotation matrix from the inverse of a full homogeneous transformation.

3.8 Additional exercises

Use metres for all position coordinates and counterclockwise-positive angles. Show your calculations and keep exact values until the final numerical step.

1. **A translated and rotated robot.** Let $p_R^W = [3 \ 1]^T$, $\theta_R^W = \pi/2$, and $B^R = [2 \ -1]^T$. Sketch the two frames and the point. Compute B^W using the affine relationship and explain why $p_R^W + B^R$ gives a different result.
2. **Recovering robot coordinates.** Suppose $p_R^W = [1 \ -2]^T$, $\theta_R^W = -\pi/2$, and $B^W = [4 \ -3]^T$. Find B^R . Substitute it into the forward relationship to verify your answer. State the frame of every vector being added.
3. **Building a homogeneous matrix.** Construct H_R^W for $p_R^W = [2 \ 0]^T$ and $\theta_R^W = \pi/4$. Use it to transform $B^R = [\sqrt{2} \ 0]^T$. Identify the physical coordinates in the output and explain the role of its last entry.
4. **World-robot-sensor composition.** Let $p_R^W = [2 \ 1]^T$, $\theta_R^W = \pi/2$, $p_S^R = [1 \ 0]^T$, and $\theta_S^R = -\pi/2$. Find $H_S^W = H_R^W H_S^R$ and use it to transform $B^S = [1 \ 2]^T$. Check your result by applying the two affine transformations separately.
5. **Inversion without a general matrix inverse.** For

$$H_R^W = \begin{bmatrix} 0 & -1 & 2 \\ 1 & 0 & 3 \\ 0 & 0 & 1 \end{bmatrix},$$

compute H_W^R using its rotation and translation blocks. Verify $H_W^R H_R^W = I_3$. Compare your result with $(H_R^W)^T$ and identify why the transpose fails.

6. **Order of operations.** Consider the two maps, expressed using a fixed coordinate system,

$$T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Compute QT and TQ and apply each to $[0 \ 0 \ 1]^T$. Explain geometrically why rotating after translating differs from translating after rotating.

7. **Distances between points.** Two points obey $B^W = p + RB^R$ and $C^W = p + RC^R$, with R orthogonal. Show that $B^W - C^W = R(B^R - C^R)$ and use this to prove that their distance is preserved. Why need the distance of a single point from its coordinate origin not be preserved?